Titanic Dataset — Exploratory Data Analysis

Objective: Use statistical summaries and visualizations to identify survival patterns, distributions, missing data, and potential anomalies in Titanic passenger data.

1. Tools and Methodology

The accompanying notebook uses Pandas for inspection and summaries, Matplotlib and Seaborn for histograms, boxplots, scatterplots, a pairplot, and a correlation heatmap. The workflow checks data types, category counts, missing values, and duplicate rows before interpreting patterns.

2. Key Findings

Survival outcome	In the commonly used 891-row Kaggle Titanic training dataset, 342 passengers survived (about 38.4%) and 549 did not (about 61.6%).
Sex and survival	Survival differed substantially by sex: approximately 74.2% of female passengers survived compared with about 18.9% of male passengers.
Passenger class	Survival rates were approximately 63.0% in first class, 47.3% in second class, and 24.2% in third class.
Age distribution	Passenger ages cover children through older adults. Age has missing entries, so summaries based on age use fewer observations unless missingness is addressed.
Fare distribution	Fare is right-skewed: most fares are relatively low, with a smaller number of high fares. Boxplots help flag these values for investigation.
Missing data	The standard training dataset has 177 missing age values and extensive missing cabin information (687 missing cabin values in the common Kaggle file). Embarked has 2 missing values.

3. Visual Analysis and Interpretation

• **Histograms:** Reveal the distribution of age and fare. Fare is usually positively skewed.

• **Boxplots:** Flag unusually high fares. Outliers should be investigated rather than automatically deleted.

• **Bar charts:** Compare survival rates by sex and passenger class. The standard dataset shows higher survival among women and first-class passengers.

• **Scatterplot:** Age versus fare, colored by survival, helps explore overlap and clustering among outcomes.

• **Pairplot:** Shows pairwise distributions and relationships among selected numeric columns.

• **Correlation heatmap:** Summarizes linear relationships among numeric variables. Class codes are ordinal, so interpret their correlations carefully.

4. Data Quality, Anomalies, and Limitations

Missing age and cabin/deck information are important limitations. High fares may be valid because ticket prices vary by class and booking arrangements. The EDA should report missingness and investigate outliers instead of assuming they are data-entry errors. Correlation and group differences describe associations; they do not prove causation.

5. Conclusion

The Titanic dataset demonstrates how EDA can uncover group differences and data-quality issues. In the common training dataset, survival was associated with sex and passenger class; fare was skewed and contained high values; and several fields were incomplete. The notebook calculates these metrics from the dataset actually loaded, so use its output to verify figures before submitting.

Dataset note: Numerical reference values above describe the commonly used Kaggle Titanic training CSV (891 rows). Seaborn's Titanic dataset has a different schema and may have slightly different missing-value counts. Always cite the dataset source and confirm the figures against your notebook output.